U1A
SAM4S16B
PA0 48
PA1 47
PA2 44 MANUAL
PA3 43 SDA
PA4 36 SCL
PA5 35 H2S_TXD
PA6 34 H2S_RXD
XIN32/PA7 32 SD_CS
XOUT32/PA8 31 SD_DETECT
PA9 30
PA10 29
PA11 28 PCS0
PA12 27 MISO
PA13 22 MOSI
PA14 21 SPCK
PA15 20
PA16 19
AD0/PA17 9
AD1/PA18 10
AD2/PA19 13
PA20 16 IO_SYNC
AD8/PA21 11 SD_MISO IO_MOSI
AD9/PA22 14 SD_MOSI IO_MISO
PA23 15 SD_CLK IO_CLK
PA24 23
PA25 25 IO_CS
PA26 26
PA27 37
PA28 38
PA29 41
PA30 42
PA31 52

D1 STATUS
D2 STATUS
R3 2k2
R4 2k2
R1 2k2
R2 2k2
+3V3
GND

MANUAL
SW2 MANUAL
GND

U1B
SAM4S16B
AD4/PB0 3
AD5/PB1 4 IO_IRQ
AD6/PB2 5
AD7/PB3 6
DAC0/PB13 59
DAC1/PB14 63

U1C
SAM4S16B
RST 39
ERASE/PB12 55
TST 40
JTAGSEL 56
TDI/PB4 3
TDO/SWO/PB5 49
TMS/SWDIO/PB6 51 SWDIO
TCK/SWCLK/PB7 53 SWCLK
DDM/PB10 56 MCU_D- 22 USB_D-
DDP/PB11 57 MCU_D+ R5 22 USB_D+
R6
XOUT/PB8 61
XIN/PB9 62

RST
+3V3
JP1 ERASE
SW1 RESET
GND

12MHz Y1
C7 27p
C8 27p
GND

U1D
SAM4S16B
ADVREF 1
VDDIN 7
VDDIO 18
VDDOUT 8
VDDCORE 12
VDDPLL 64
GND 2
+3V3
FB1 FerriteBead
C9 C5 C4 C1 100n
C10 C6 C3 C2 100n
R7 1
C11 100n
C12 4u7
GND

J13
Micro SD
DAT2 1
DAT3/CD 2
CMD 3
VDD 4
CLK 5
VSS 6
DAT0 7
DAT1 8
DET_A 10
DET_B 9
SHIELD 11
SD_CS
SD_MOSI
SD_CLK
SD_MISO
SD_DETECT
+3V3
C17 1u
GND

U3
TLV70233_SOT23-5
+5V
+3V3
IN 1
EN 3
OUT 5
GND 2
C19 1u
C20 1u
GND
Should add power switch/button

This is actually a 9-pin part (symbol is wrong)
J4
SPEC H2S
H2S_RXD 1
H2S_TXD 2
3
4
5
GND 6
7
+3V3 8

I2C: 0x25
J6
SDP810
SCL 1
+3V3 2
GND 3
SDA 4

I2C: 0x40
J5
SFM3000
SDA 1
GND 2
+5V 3
SCL 4

J12
SERIAL
+5V
+5V
IO_IRQ 3
IO_MISO 5
IO_SYNC 7
9
1 2 IO_CS
4 IO_MOSI
6 IO_CLK
8
10
GND
GND

C18 4u7
+5V
GND
J2
USB
VBUS 1
USB_D+ 3 D+
USB_D- 2 D-
ID 4
GND 5
Shield 6
GND

J7
EAR
T
S
J9
Ear
1
2
GND

+5V
R8 10k
C21 1u
R10 1M
R11 1M
GND
R14 is actually 2K on board
C23 1u
GND
+5V
C27 100n
GND
U4
MAX4466
R14 20k
R16 200k
C25 100p
GND

Audio jack wiring error:
shield and tip need to be swapped
These sockets also have ring connections

J8
MASK
T
S
J10
Mask
1
2
GND

+5V
R9 10k
C22 1u
R12 1M
R13 1M
GND
C24 1u
GND
+5V
C28 100n
GND
U5
MAX4466
R15 20k
R17 200k
C26 100p
GND

U2
MAX1167
AVDD 13
DVDD 16
DOUT 1 MISO
SCLK 2 SPCK
DIN 3 MOSI
CS 14 PCS0
AIN0 5
AIN1 6
AIN2 7
AIN3 8
EOC 4
REF 9
REFCAP 10
DGND 15
AGND 12
GND
C16 100n
+5V
+3V3
C15 100n
GND
ADCREF
C14 100n
C13 10u
GND
GND
GND
J11
ADCREF
1
2

Not connected on PCB!
+3V3
J3
STDC
RST 12 RST
VCC 3
SWCLK 6 JCLK/SWCLK
SWDIO 4 JTMS/SWDIO
8 JTDO/SWO
10 JTDI/NC
9 JRCLK/NC
13 VCP_RX
14 VCP_TX
GND 5
GNDDetect 11
GND

J1
SWD
RST 1
3
5
GND 7
+3V3 9
2 SWCLK
4 SWDIO
6 GND
8 +3V3
10

H1 H2 H3 H4
GND

Sheet: /
File: instrumentation.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. kicad 6.0.2-378541a8eb-116~ubuntu20.04.1
Id: 1/1